Repository Structure

- Data/
- ANN/
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Prerequisites

Required Python Packages

pip install pandas numpy matplotlib seaborn scikit-learn shap torch

note: you must also install torch for your system.

Dataset Requirements

The following datasets should be in the Data/ folder:

- Telco-Customer-Churn.csv (for ANN)
- diabetes.csv (for Logistic Regression)
- AirQualityUCI.csv (for Isolation Forest)
- spambase.data (for KNN) - should be downloaded and placed **in KNN/** folder

1. Artificial Neural Network (ANN)

Steps to run:

1. Run the data_preprocessing.ipynb python notebook
2. This will generate a processed_customer_churn.csv file
3. Run the ann_implementation.ipynb Cell by cell to train the model and visualize outputs

2. Isolation Forest

Steps to run:

1. Run the data_preprocessing.ipynb python notebook
2. This will generate csv files for the processed data in the Data directory
3. Run the isolation_forest.ipynb Cell by cell to train the model and visualize outputs

3. K-Nearest Neighbors (KNN)

Steps to run:

1. Run the preprocessing.ipynb python notebook
2. This will generate spambase_processed_data.csv in the Data directory
3. Run the analysis.ipynb Cell by cell to train the model and visualize outputs

4. Logistic Regression

Steps to run:

1. Run the data_preprocessing.ipynb notebook
2. This will generate processed data files
3. Run logistic_regression.ipynb cell by cell and train the model, then visualize results

NOTE: SHAP ANALYSIS CELLS MAY TAKE LONG TO RUN. ONE MAY ADJUST NUM SAMPLES TO TACKLE THIS